RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

INNOVATIVE TEACHING PLAN

Year/Semester/Section: III/V/A **Date: 30/09/2024**

Topic : Recursive neural networks with human brain **CO Coherency: CO3**

Duration : 45 minutes

Recursive Neural Networks (RNNs) & Human Brain Concept-Mind Mapping

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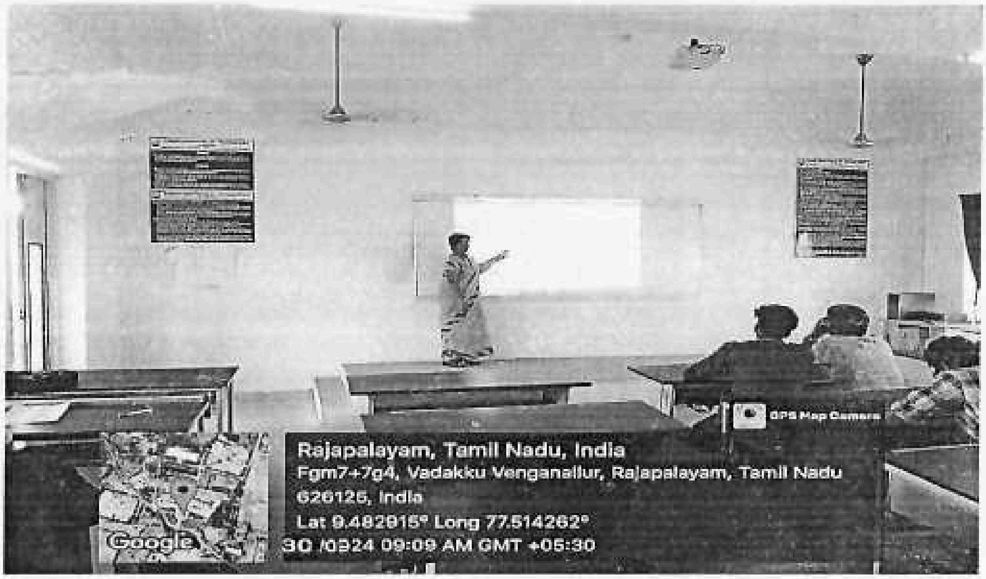
• A mind map is a visual representation that shows concepts and ideas. It's a technique for visual thinking that helps with data arrangement.

• It swiftly jots down ideas. So, this practice will help students recognize the theoretical ideas in the image and facilitate quick recall for their tests.



1. **Definition of RNNs**

   Structure: RNNs process data sequences, using previous outputs as inputs for the next steps.

   Function: Ideal for tasks involving sequential data (e.g., language processing, time series analysis.

2. **Human Brain Analogy**

   Neural Structure: Neurons in the brain transmit information in a recursive manner, similar to RNNs.

   Memory & Learning: Just as RNNs remember past inputs, the human brain uses previous experiences to inform future actions.

3. **Applications**

   Natural Language Processing (NLP): Understanding and generating human language.

   Image Recognition: Analyzing sequences of images or patterns.

4. **Similarities to Brain Function**

   Hierarchical Processing: RNNs process information in layers, akin to the brain's hierarchical processing of sensory information.

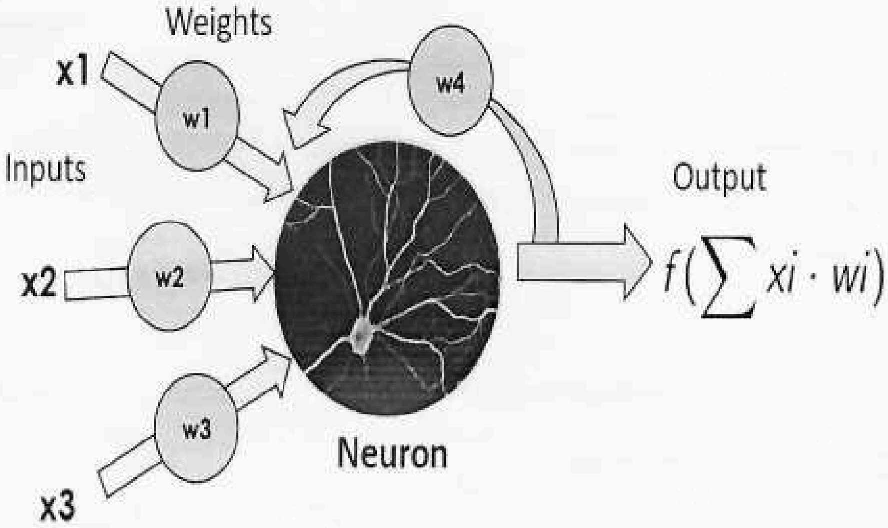
   Adaptability: Both RNNs and the brain adapt based on new information (neuroplasticity).

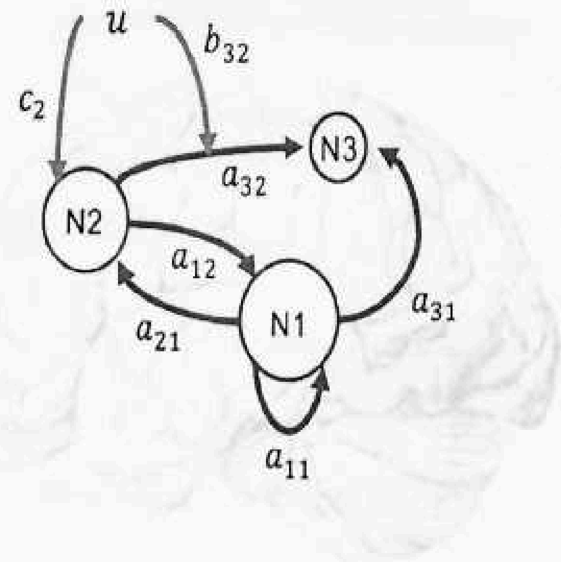
5. **Limitations**

   Vanishing Gradient Problem: RNNs can struggle with long-term dependencies.

   Brain Complexity: The human brain is far more complex and efficient in processing information.

6. **Future Directions**
   - **Enhanced Models:** Developing more sophisticated neural networks inspired by brain functionality.
   - **Interdisciplinary Research:** Collaborating across neuroscience and AI to improve RNNs.





$$A = \begin{bmatrix} a_{11} & a_{12} & 0 \\ a_{21} & 0 & 0 \\ a_{31} & a_{32} & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & b_{32} & 0 \end{bmatrix}$$

$$C = \begin{bmatrix} 0 \\ c_2 \\ 0 \end{bmatrix}$$

Benefits:

By mapping out the relationship between RNNs and the human brain, students can develop a deeper understanding of both fields. This approach not only enhances their knowledge but also encourages interdisciplinary thinking, creativity, and critical analysis, preparing them for future challenges in technology and science.

Learning Outcomes

- Deeper Understanding: Mind Mapping approach allowed students to grasp complex concepts better, as they had more time to review the materials at their own pace before class.

- Feedback for Improvement: Valuable feedback from students highlighted areas for improvement, such as clearer instructions for pre-class preparation and more diversified in- class activities

Feedback Questions

1.Did you find mind map clear and easy to follow?

O Yes O No

2.Did the mind map help you visualize your ideas and thoughts more clearly?

O Yes O No

Was there enough flexibility in how you could organize or add ideas?

O Yes O No

Did the mind map make it easier to understand the relationships between different ideas or concepts?

O Yes O No

What improvements would you suggest to make the mind mapping process better or more effective?

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Department of Artificial Intelligence and Data Science

Academic Year 2024 – 2025 (Odd)

Degree, Semester & Branch: BTech/V/AD-A

Course Code & Title:AD3501-Deep Learning

Name of the Faculty member (s): Mrs.R M Rajeshwari

Innovative Practice Description

- **Unit / Topic: Unit III- RNN with Human Brain**
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO6
- **Activity Chosen: Mind Mapping**
- **Justification:**
    - ➢ Using mind mapping to compare RNNs with the human brain helps to visually and conceptually represent how both systems process information, store memory, learn, and adapt.
    - ➢ It serves as an effective tool for understanding the similarities and differences between biological and artificial neural networks, and can also highlight areas where AI systems, like RNNs, may still fall short in replicating the full complexity of human cognitive processes.

Time Allotted for the Activity: 45 Minutes

- **Details of the Implementation:**
- The comparison between the human brain and Recurrent Neural Networks (RNNs) using the mind mapping method is primarily implemented as a conceptual framework to understand the similarities and differences between biological and artificial intelligence. The process begins with creating a mind map that centers on the theme "Comparison of Human Brain and RNN." Key branches are then created for major areas such as information processing, memory, learning, feedback loops, and real-time adaptation, with sub-branches detailing specific aspects of each system, such as short-term memory in the brain versus hidden states in RNNs. Each component is mapped to illustrate how both the human brain and RNNs function in real-time. For example, in terms of memory, the brain utilizes the hippocampus and cortex for short-term and long-term memory, while RNNs rely on hidden states to retain information over time. The brain learns through neuroplasticity, continuously adapting with experience, while RNNs adjust weights through backpropagation during training. Feedback loops in the brain are dynamic, responding to real-world sensory inputs, whereas RNNs use feedback within the network by passing previous outputs into the system.. Thus mind maps allow the students to visualize how RNNs emulate cognitive functions of the brain, such as pattern recognition and contextual learning, while also highlighting areas where the brain excels, such as generalization and abstract thinking

CO – PO / PSO mapping: (... 2 – Moderate 3 – High)

- **PO / PSO mapped:**

CO	PO 1	PO2	PO3	PO9	PO10
CO4	2	2	2	3	3

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Innovative practice	PSO1	PSO2	PSO3	-	-
	2	2	2	-	-
Justification for correlation	**PSO1**:The context of software development. RNNs are used to process sequential data, such as time series, text, and speech. This aligns with PSO1 by highlighting how convolution techniques are applied to data in software development tasks, especially in domains like speech recognition, natural language processing, and time series forecasting. **PSO2**:RNNs are a core architecture for representing and analyzing time series data. They handle dependencies between time steps through internal loops and pass information through time steps, making them ideal for tasks such as forecasting, anomaly detection, and sequence prediction. PSO2 is reflected in the knowledge of how to represent and analyze time series data using RNNs. **PSO3**: RNNs are applied to real-world problems such as machine translation, speech recognition, and financial forecasting. By using techniques like convolutions (in combination with RNNs, particularly in architectures like CNN-RNN hybrids), these models can extract features from sequences and make predictions that solve practical, real-world issues.				

- **Images / Screenshot of the practice: Report Attached**
- **Reflective Critique:**
- ***Feedback of practice from students and other stakeholders:*** **Separately Attached**

  - ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice) **Report Attached**

  - ***Challenges faced in implementation:***

    While the mind mapping method is a valuable tool for visualizing and comparing the human brain and RNNs, it comes with several challenges due to the complexity of both systems and the limitations of artificial neural networks in mimicking the full range of human cognition. These challenges include oversimplification, issues with representing real-time learning, difficulties in mapping memory systems, and cross-disciplinary complexities, all of which need to be carefully addressed for an accurate and meaningful comparison.

Signature of Faculty Member **HOD**